

Project and PI Information:

Project Title: Teaching Large Language Models to Explore in Reasoning

PI: Chelsea Finn

Group member SUNet IDs: ALL

Abstract:

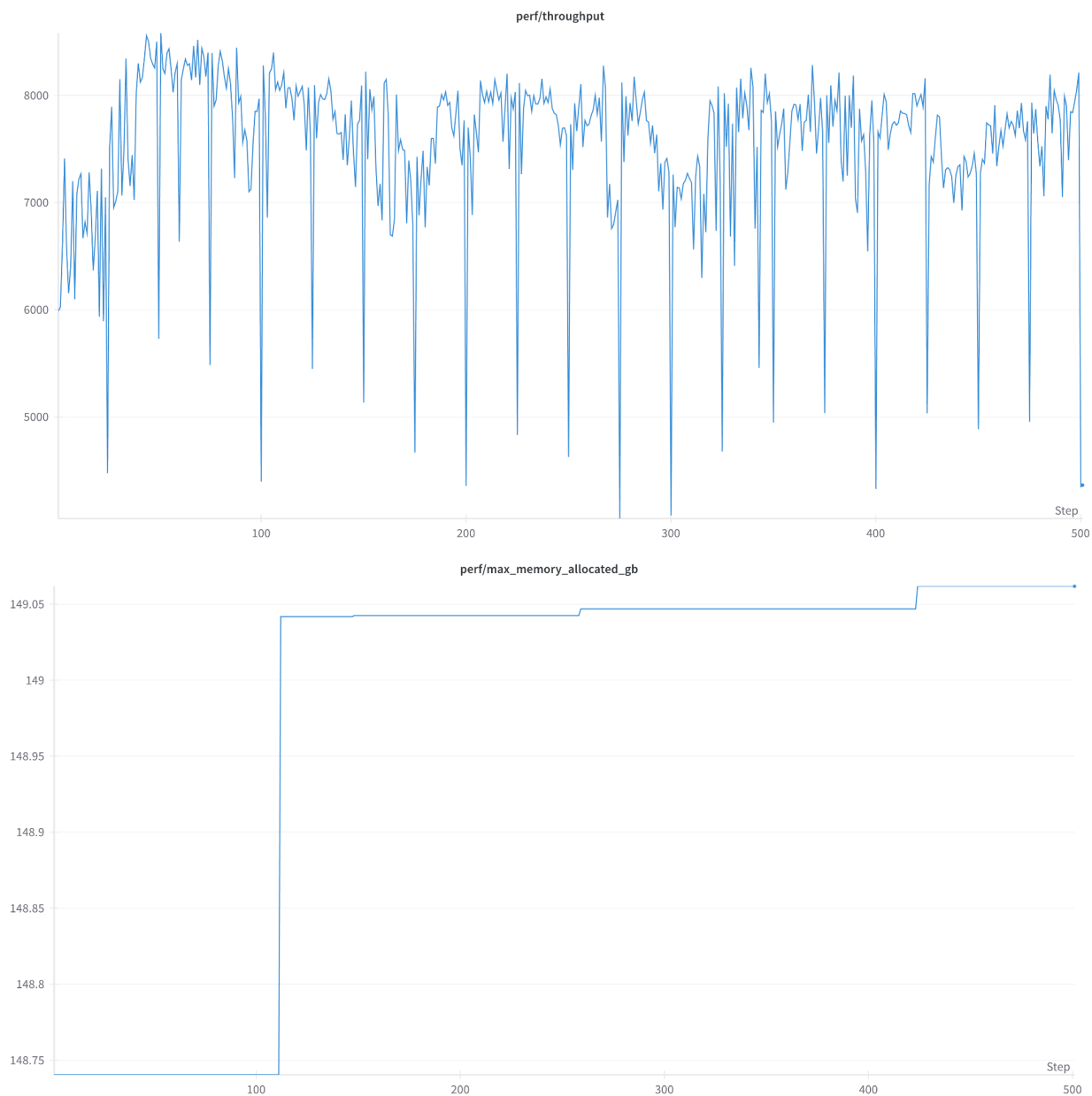
Foundation models have made remarkable strides across numerous domains, including language, mathematics, and robotics. A crucial driver of this advancement is their reasoning capabilities, exemplified by OpenAI's o3, Google DeepMind's Flash Thinking, and Deepseek R1. Reasoning allows models to iteratively refine their solutions by intelligently exploring possibilities rather than simply recalling memorized patterns. Reinforcement Learning (RL) has emerged as a dominant paradigm for enhancing reasoning, enabling models to explore solution spaces and iteratively improve through reward-driven learning. We aim to investigate and develop algorithmic ideas in RL to make reasoning more effective for foundation models.

Scientific Overview:

Online reinforcement learning (RL) has been widely adopted as a scalable method for teaching large language models (LLMs) to tackle problems with verifiable solutions, like mathematical problem solving and coding [1, 3]. This looks like fine-tuning large pre-trained models using verifier reward models and algorithms like proximal policy optimization (PPO) or group relative policy optimization (GRPO). However, recent works have shown that such reinforcement learning fine-tuning does not help these models learn new reasoning capabilities. Instead, this fine-tuning sharpens the pre-trained distribution around behaviors that achieved higher rewards in the reinforcement learning phase. Such refinement helps models tackle problems already seen efficiently (enabling higher pass@1 success rate) but reduces coverage over problems that are solvable [2]. Additionally, such fine-tuning reduces creativity and the ability to model randomness or exhibit creativity [4]. This project aims to train large language models to learn multiple methods for solving a given problem, providing the model with a larger repertoire of techniques required to solve real-world problems. We approach this problem through learning to explore creative hypotheses while preserving the ability to carry out efficient reasoning conditioned on any one hypothesis.

Our team has significant experience working with SLURM-managed compute clusters, having completed multiple projects on both the SC cluster and Marlowe. We have also effectively operated in various other heterogeneous computing environments. One aspect of higher computational requirements include scaling large model parameters (going from 4b to 32b). Additionally, our algorithms frequently require multiple models concurrently held in GPU memory, either co-located across multiple GPUs or distributed across independent GPUs. Through the course of RL finetuning, these models must routinely communicate intermediate results to make forward progress. This high communication regime, along with the need for multiple GPUs, underscores our need for a reliable, high-performance cluster, which Marlowe specifically offers.

We have demonstrated the high performance and efficient GPU utilization of our implementations. Empirical evidence from profiling (shown below) indicates our language model experiments consistently achieve high throughput, regularly exceeding 8000 tokens per second. Additionally, GPU memory usage profiling shows effective saturation of the available ~150GB VRAM on the H200 GPU. These metrics reflect our implementations' ability to fully leverage the capabilities of given GPUs, making optimal use of the hardware provided by Marlowe. We additionally add an MFU plot as a function of training steps, though the MFU reported is fairly low, this is a known downside of the finetuning regime that we operate in.





Our typical computational jobs require **8-32 NVIDIA H100 GPUs each**, with **GPU utilization consistently maintained between 80-100%**, and minimal CPU demands (typically under 3% on AMD EPYC 7742 64-core CPUs). Jobs are executed on both single-node and multi-node configurations, usually employing **one to four nodes per job**. RAM usage typically ranges from 20GB to 64GB under standard conditions, but can scale to several hundred gigabytes when model parameter offloading becomes necessary. Additionally, as described previously, our workflows require substantial inter-GPU communication, benefiting significantly from Marlowe's NVLink-enabled GPUs, which allow fast communication between GPUs. Furthermore, our training workflows are checkpoint-enabled, allowing job pausing and restarts, with each checkpoint occupying approximately 8–16GB of storage.

Our rationale for leveraging Marlowe stems from the inadequacy of available A100/H100 GPUs through alternative sources such as Sherlock, Google Colab, and our lab server, since they often suffer from high contention. Our work would greatly benefit from the stable and consistent availability of GPUs provided by Marlowe. The dedicated GPU resources and priority scheduling on Marlowe would allow us to conduct our GPU-bound experiments with faster turnaround, significantly accelerating our research cycles. While this same argument can be applied to persistent GPU offerings from remote cloud providers, monetary constraints prevent us from provisioning GPUs through these means.

Computational Profile:

Most of our experiments involve fine-tuning open-source language models with 3–8 billion parameters on jobs utilizing **8-32 NVIDIA H100 GPUs**. **A single experiment runs for 12-36 hours** in a primarily compute-bound workload. Since our research uses RL frameworks such as VeRL (<https://github.com/volcengine/verl>), TRL (<https://github.com/huggingface/trl>), and OpenR1 (<https://github.com/huggingface/open-r1>) for single and multi-node training of foundation models, we inherit many of the GPU-side throughput optimizations done by these open source RL finetuning implementations, along with benefits of these being active open-source implementations that keep having efficiency updates.

Technical Requirements & Feasibility:

Our software requirements are minimal, given that all the code utilized in our workflows is open-sourced and pip-installable. A standard Linux kernel environment with an up-to-date Python installation and compatible NVIDIA GPU drivers suffices for our needs.

Storage & Data:

Our storage needs primarily consist of scratch space for intermediate files, checkpoints, and temporary data generated during training runs. We estimate our typical scratch storage requirement to range from 400-500 GB per experiment, depending on checkpoint frequency. Data for experiments will be sourced from publicly available datasets, directly downloaded onto the cluster storage, or transferred via secure file transfer protocols such as SCP or rsync. We regularly checkpoint our models during training, with each checkpoint occupying approximately 32 GB. We also routinely clean up experiments post-analysis and retain only critical checkpoints and logs. Thus, an overall estimation of required scratch space would be about 200 TB.

Impacts and Acknowledgements

This project directly contributes to the broader goal of making large language models better reasoners. We plan to publish our research in a leading machine learning venue, specifically NeurIPS 2026. We also plan to open-source our software to enable other groups to replicate, adapt, and build upon our methods.

References

[1] Snell, C., Lee, J., Xu, K., & Kumar, A. (2024). Scaling LLM test-time compute optimally can be more effective than scaling model parameters. arXiv.
<https://arxiv.org/abs/2408.03314>[\]\(https://arxiv.org/abs/2408.03314\]](https://arxiv.org/abs/2408.03314)

[2] Yue, Y., Chen, Z., Lu, R., Zhao, A., Wang, Z., Yue, Y., Song, S., & Huang, G. (2025). Does reinforcement learning really incentivize reasoning capacity in LLMs beyond the base model? arXiv. <https://arxiv.org/abs/2504.13837>

[3] Havrilla, A., Du, Y., Raparthy, S. C., Nalmpantis, C., Dwivedi-Yu, J., Zhuravinskiy, M., Hambro, E., Sukhbaatar, S., & Raileanu, R. (2024). Teaching large language models to reason with reinforcement learning. arXiv. <https://arxiv.org/abs/2403.04642>

[4] West, P., & Potts, C. (2025). Base models beat aligned models at randomness and creativity. arXiv. <https://arxiv.org/abs/2505.00047>

Survey Responses

PI Name *

Chelsea Finn

Group SUNet IDs *

Provide SUNet IDs that require access to this project.

Write "ALL" to grant access to your entire Basic Access/preemptive group. Otherwise, list specific SUNets who should have access to this project in a comma-separated list.

ALL

Project Title *

Project Title—10 words max

Teaching Large Language Models to Explore ir

Project Description *

Abstract—summarize the scientific/engineering goals, anticipated impact, and significance. 250 words max.

Foundation models have made remarkable strides across numerous domains, including language, mathematics, and robotics. A crucial driver of this advancement is their reasoning capabilities, exemplified by OpenAI's o3, Google DeepMind's Flash Thinking, and Deepseek R1. Reasoning allows models to iteratively refine their solutions by intelligently exploring possibilities rather than simply recalling memorized patterns. Reinforcement Learning (RL) has emerged as a dominant paradigm for enhancing reasoning, enabling models to explore solution spaces and iteratively improve through reward-driven learning. We aim to investigate and develop algorithmic ideas in RL to make reasoning more effective for foundation models.

PTA *

Please enter the PTA account to charge for this project.

Note: Usage currently is free. We will provide advance notice before chargeback begins.

--- Technical Requirements ---

The following questions are common to all projects regardless of scale.

--- Section 3: Computational Profile ---

This section captures your typical workload profile (nodes/GPUs, wall-time, concurrency) so that we can assess fit and plan system capacity. To make this determination we include scaling considerations (e.g., number of nodes/GPUs, wall time).

Job Type *

Most Typical Job Type

Model Development/Training ▼

Wall Time *

Typical Wall Time per Job

>24 hours ▼

GPUs per Node *

For a typical job how many GPUs per Node will be required?

8 ▼

NodeCount Typical Job *

Enter how many nodes your typical job for this project will require.

2 _____

NodeCount Max Job *

Enter how many nodes your largest job for this project will require.

8 _____

Continue to Technical Requirements

--- Section 4: Technical Requirements & Feasibility ---

In this section, please provide detail about software frameworks, containerization, and any special system needs.

Software *

Required Software Frameworks/Libraries. Please be specific about any required versions.

Cuda 12, NVCC

Containerization *

Required Containerization Tools, e.g., Singularity/Apptainer, Conda, OnDemand, etc.

Docker, Conda

Special Configuration *

Do you have custom system configuration needs (e.g., intensive I/O, special kernel modules)?

☐ Yes

☒ No

Special Configuration Details

If yes, briefly describe those needs

Your answer

Support Needs *

Do you expect to need support from Marlowe staff? For example, help with setting up your

Continue to Storage Requirements

--- Section 5: Storage/Data Requirements ---

In this section, please describe storage requirements, data sizes, data movement.

Storage Allocation Note

Project-specific scratch storage is available and may be requested below.

Note: Project-specific /projects storage is not available for Medium or Large projects, existing data in /projects remains accessible.

Marlowe is not intended for long-term data retention. To ensure fair access, storage must remain fluid and responsive to operational needs.

Scratch Storage *

How many TB of scratch storage does this project require? (Mounted on /scratch, NOT backed up or snapshotted) Enter a value between 0 and 500 TB.

200

Primary Data Source *

Where will the bulk of your input data come from?

Fetches from the internet or cloud (e.g., S3, GCP, HuggingFace) ▾

Primary Data Source Details

Optional: Only complete this if you selected 'Other' above.

Mostly from huggingface (to/from)

Estimated Data Movement

Roughly estimate how much data will move **into** and **out of** Marlowe during the course of the project. Mention formats (e.g., RAM, Parquet, checkpoint files) and tools.

Continue to Impact and Acknowledgments

--- Section 6: Impact & Acknowledgments ---

In this section, please specify expected outcomes (e.g., publications, models, datasets), acknowledge data restrictions, and confirm that you will cite Marlowe when it results in publications/presentations.

Anticipated Outcomes *

E.g., publications, open-source software, domain-specific breakthroughs, etc.

We anticipate this research will result in several publications in top machine learning conferences or journals, with code for each publication being made open-source.

Risk Acknowledgement *

Marlowe is approved for low and medium-risk data only. Please confirm that your group will use only low and medium-risk data and NOT use high-risk data (e.g., no PHI, PII, etc.) on Marlowe.

☒ I confirm that my group will only use low and medium-risk data on Marlowe.

Citations *

To track Marlowe's impact on Stanford research, VPDOR/SDS requires users to cite Marlowe in publications or presentations arising from its use. Please confirm your agreement and use the following citation: <https://doi.org/10.5281/zenodo.14751899>

☒ I agree to acknowledge/cite Marlowe in any publications or presentations resulting from the use of Marlowe.

Need Help?

If you have questions about the project application requirements, please contact marlowe-info@stanford.edu.